

E-Skin Report Winter 2026

By Cole Abbott

Summary

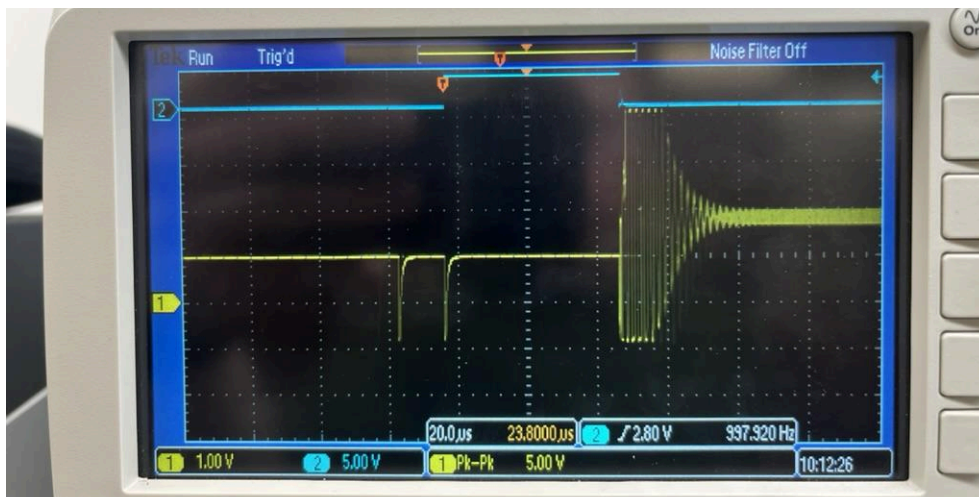
This winter I continued working on the E-Skin project under Shihan Lu. Over the summer and Fall quarter, a PCB had been designed, ordered, and assembled by Shihan and some other students. However, the USB was not functional, the PIC was crashing due to voltage spikes, and there was a lot of signal interference on the PCB board. The majority of the quarter was spent diagnosing these issues. Towards the end of the quarter, we were able to collect some data from the E-Skin.

PIC USB Issues

The USB peripheral of the PIC32 was not working on the PCB board. After comparing the auto generated configuration code for the dev board and the PCB board, I saw that one difference was the clock configuration. After checking the datasheet, I learned that a 12Mhz or 24Mhz Primary oscillator is needed in order to use USB on the PIC. The PCB was designed with a 26Mhz crystal, but a 12Mhz could be swapped in with no modification to the board. After reworking the board, the USB functionality works.

PIC Crashing

Once the USB started working, ADC data could be collected from the PIC. This worked while the analog portion of the circuit was disabled, but upon powering it, the PIC would crash, and had to be reset before it was able to send data again. We eventually determined that the reason for this crash was a negative voltage pulse going into the ADC.



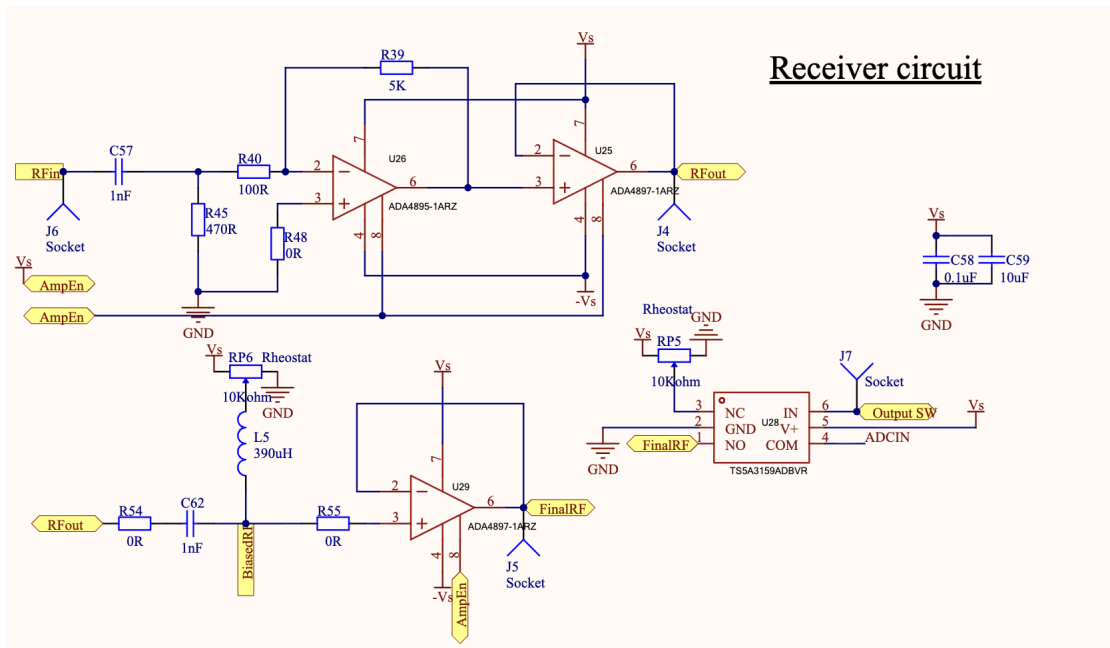
Negative voltage on ADCIN signal

As seen above, there were 2 -1v pulses, and then a signal with oscillations going to -1v reaching the PIC. The first 2 pulses are from the pulse generator charging up, and the negative signal is due to how the signal is amplified by the receiver circuit. The receiver circuit does have an adjustable bias tee to shift the signal up, but it is not able to shift the signal high enough to keep it positive.

To further debug this issue, we tried connecting the receiver circuit of the PCB to the ADC of the dev board. The PIC crashed in the same way, but now we could add additional components into the circuit to test if we could remove the negative spikes. First, we tried to use a diode to clamp the voltage to be positive by connecting the diode from the signal to ground, but this did not work. Placing the diode in series with the signal did attenuate the negative voltage enough for the PIC to not crash, however this did affect the quality of the received signal. We also tried connecting a resistor in series with the signal, and got much better signal quality. High resistances of $>100K$ did attenuate the signal significantly, and low resistances of below 100Ohms did not prevent the PIC crashing. Values of 1k-3.3k gave a good signal and the PIC could function reliably. I believe the reason this works is because the resistor limits the current that can flow into the PIC (or rather out of due to the negative voltage) and then the built in protection circuitry in the PIC is not overwhelmed.

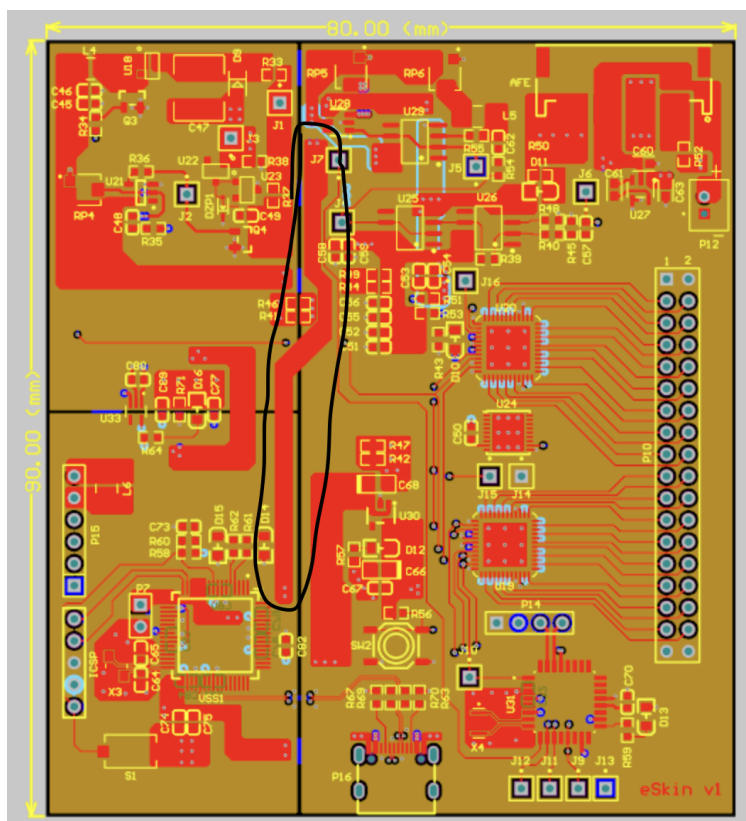
Next, we wanted to add a similar resistor setup to the PCB board. We could not simply add a component in series to a PCB trace, so instead we removed a component, the analog switch, that was not necessary, and then could add a resistor in its place. However, this did not work, and the PIC still crashed, in fact it still crashed when the resistor was removed, and there was no connection between the ADC and the receiver circuit. This was indicative of significant interference on the PCB between the ADCIN trace and other signals on the board, mainly the HV pulse.

The ADCIN trace runs past the pulse generator, and picks up a lot of interference, enough to crash the PIC. To test this theory, we cut the ADCIN trace, so that portion that is near the pulse generator is not attached to the ADC. The cut trace is bypassed by a wire with a resistor in series. This worked, and the PCB was now functional with these modifications, and we could move on to collecting data.



Receiver Schematic

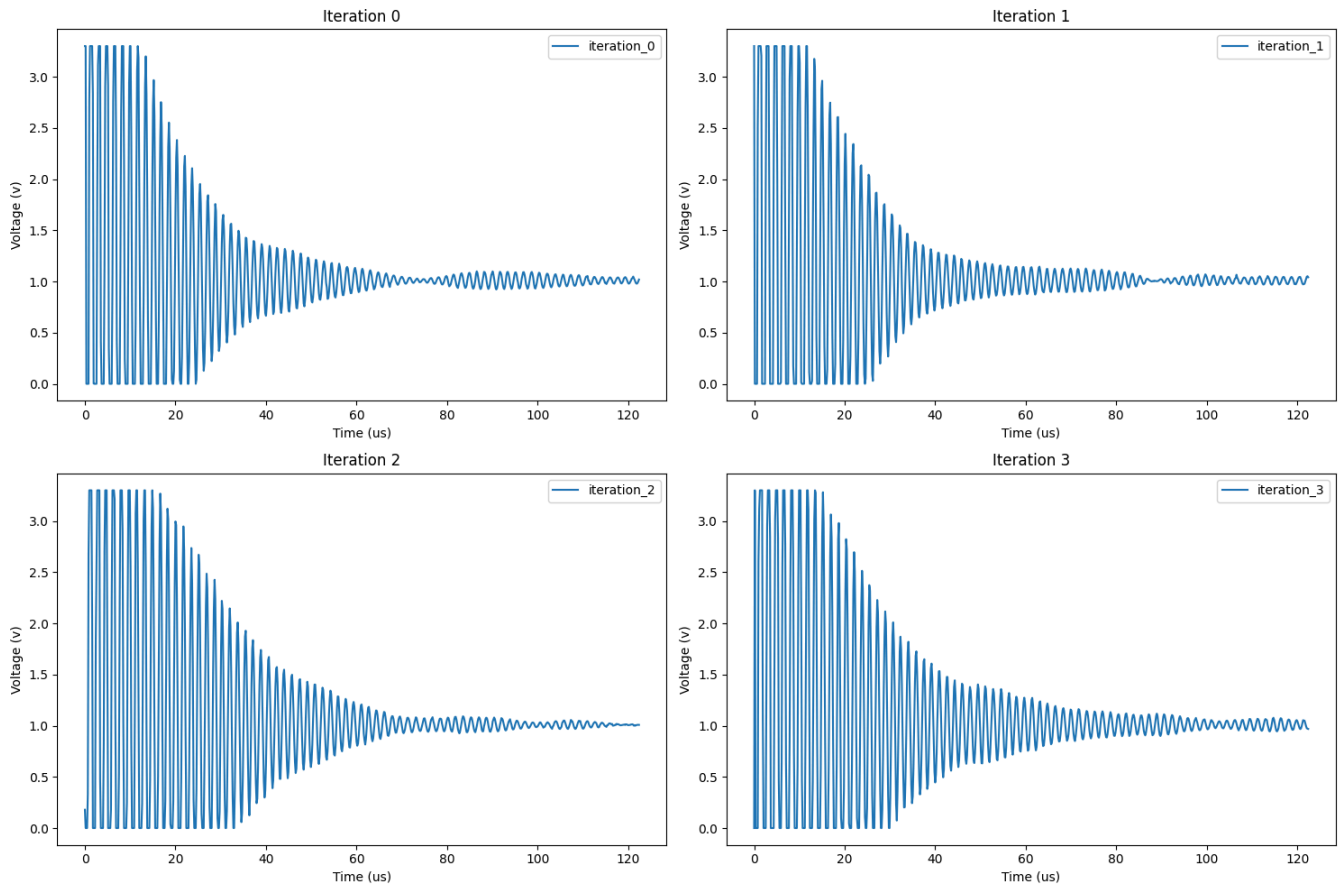
The removed switch ,U28, connects the output of the receiver, Final RF, to the PIC ADC, ADCIN.



PCB Layout, ADCIN trace circled in black

Data Collection

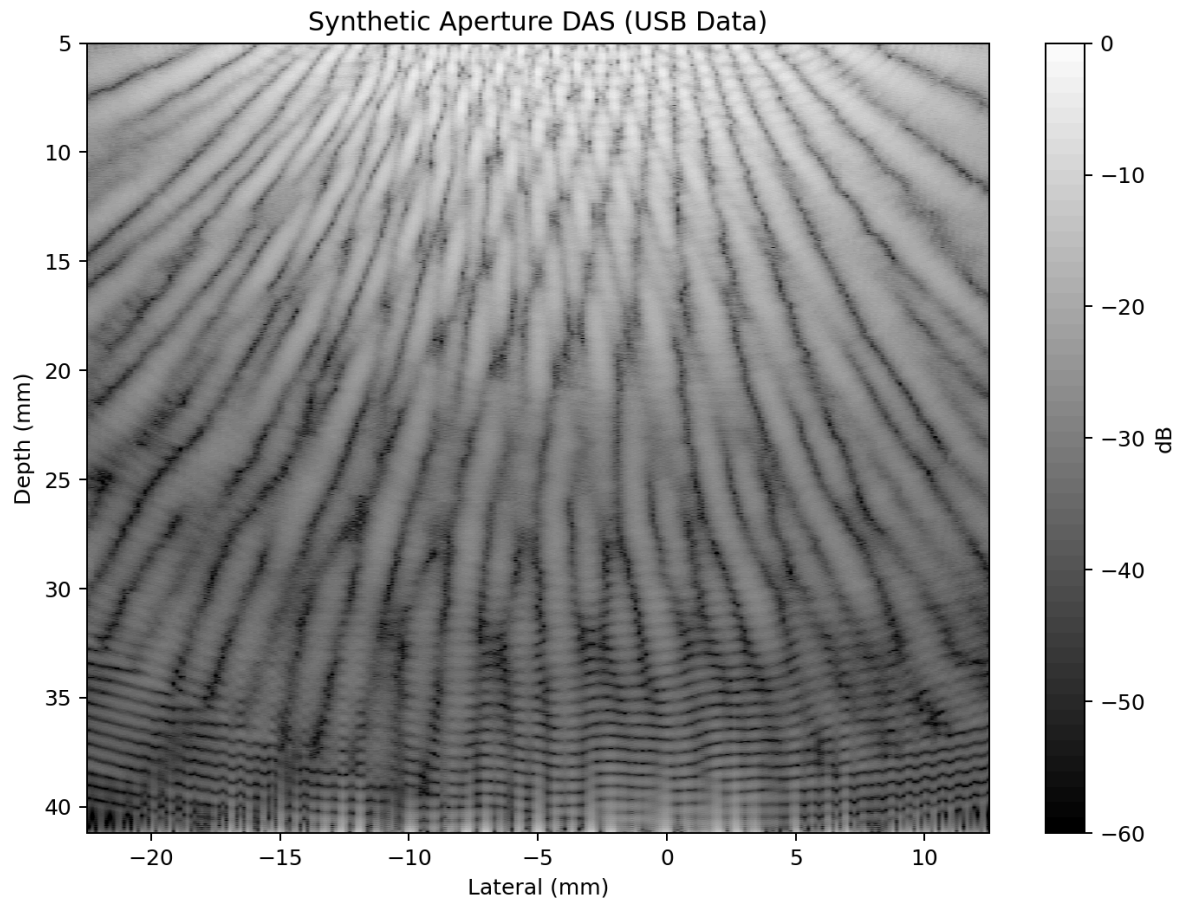
With the PCB functional, we can now collect data from a transducer array.



Data collected from transducer array

We were able to collect data from a 4 channel transducer array, and save it to a file over usb, as well as visualize it in real time. This validates that the circuit and data collection is all functioning properly.

Now we can run this through the beamforming script, to determine where force is being applied to the skin. This will combine the raw data from all the channels into an image that shows where force is applied to the skin. However, the quality of the data was not good enough for this to work properly. With a better transducer setup, (right now we are just pressing it by hand against the skin), and better filtering, I think we can get a much better result



Beamforming script output with real data

Next Steps

Next quarter, we will refine the data collection and analysis process, and integrate with a modified 3d printer and can repeatedly press on the skin to evaluate the performance of the e-skin. Shihan is also redesigning the PCB board to remove the signal interference issues.